

# Kernel-checked constraints from DESI DR2 BAO in Lean 4: certified exclusions of $\Omega_m$ bands and of $H_0 \geq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ with a BBN prior

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## Abstract

We turn two standard cosmological statements into theorems checked by the Lean 4 kernel, with solvers and minimisers kept outside the trusted base. For flat  $\Lambda$ CDM with radiation neglected and the Gaussian DESI DR2 BAO likelihood [1]: (i) the BAO  $\chi^2$  exceeds 35.2897 for every  $\Omega_m \in [0, 0.255] \cup [0.35, 1]$  and every value of the BAO scale  $hr_d$ , while it is at most 10.2897 at a specific point, so these bands are excluded at  $\Delta\chi^2 > 25$  with  $hr_d$  profiled ( $5\sigma$  for one parameter of interest under Wilks’ theorem; about  $4.6\sigma$  if read as a joint two-parameter region); Einstein–de Sitter has  $\chi^2 \geq 1415.82$  at every  $hr_d$ . (ii) With the BBN prior  $\omega_b = 0.02218 \pm 0.00055$  (DESI DR2 eq. 14 [1, 2]) and the Aubourg et al. fitting formula for the sound horizon [3], the combined  $\chi^2$  exceeds  $10.2897 + 25$  for every  $\Omega_m \in [0, 1]$ , every real  $\omega_b$  and every  $h \geq 0.73$ , while it is at most 10.2897 at  $h \approx 0.685901$ : a certified, conservative form of the known tension between BAO+BBN and the distance-ladder value  $H_0 \approx 73$  [4]. The proofs use a rational interval evaluator with a soundness theorem, monotonicity in the parameters, bounds of log, exp and real powers derived from Mathlib’s series estimates, and integer certificates closed by `decide +kernel`; only the axioms `propext`, `Classical.choice` and `Quot.sound` are used. The preregistered slab layout of (i) failed and a disclosed post-hoc refinement was needed. These are certified versions of results already known numerically; there is no new physics here.

## 1 Why certify a likelihood statement

Statements such as “ $\Omega_m > 0.35$  is excluded” or “BAO+BBN disfavour  $H_0 = 73$ ” are normally established by floating-point minimisation over nuisance parameters. A kernel-checked version removes the integrator, the minimiser and the floating-point arithmetic from the trusted base: if the theorem compiles, it holds for the stated model and data for *every* value of the profiled parameters, not only those a minimiser visited. This continues a pilot that certified BAO distances only [5]. The model and the likelihood remain inputs that a reader must accept or not; the theorems certify the computation, not the cosmology.

## 2 Model, data and statements

$E(z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$  (flat  $\Lambda$ CDM, radiation off),  $\chi(z) = \int_0^z dz'/E$ . The DESI DR2 BAO data vector (13 measurements at 7 effective redshifts,  $z = 0.295, \dots, 2.33$ ) and its block-diagonal covariance  $C$  are read as exact rationals;  $P = C^{-1}$  is computed exactly and the identity  $PC = I$  over the full  $13 \times 13$  matrices is checked by the kernel. Predictions are  $K\chi$ ,  $K/E$  and  $K(z\chi^2/E)^{1/3}$  for  $D_M/r_d$ ,  $D_H/r_d$ ,  $D_V/r_d$ , with  $K = c/(100 hr_d)$ . For (ii),  $r_d = 55.154 e^{-72.3(\omega_b + 0.0006)^2 / (\omega_{cb}^\alpha \omega_b^\beta)} \text{ Mpc}$

with  $\alpha = 0.25351$ ,  $\beta = 0.12807$  [3],  $\omega_\nu = 0.0107 \times 0.06$  and  $\omega_{cb} = \Omega_m h^2 - \omega_\nu$ , exactly as in an earlier floating-point analysis by the author (repository `AutoevolveAI`, `results/bao_bbn_h0`, with radiation included, which found  $H_0 = 68.57 \pm 0.60$ ); we import its Lean skeleton (`formal/ANSE/BAO_BBN_H0.lean`, monotonicity of  $r_d$  and of  $hr_d$ ) byte for byte. For comparison, DESI’s own BAO+BBN analysis gives  $H_0 = 68.51 \pm 0.58$  [1]. Table 1 lists every preregistered statement and its verdict.

| id       | statement<br>(kernel-checked<br>noted)                                                                    | unless | Lean                                | verdict              |
|----------|-----------------------------------------------------------------------------------------------------------|--------|-------------------------------------|----------------------|
| P2 / H6  | $\chi_{\text{BAO}}^2(0.29743, K_{\text{fit}})$<br>[10.2525, 10.2897]                                      | $\in$  | <code>P2.Fit.chi2_fit</code>        | yes                  |
| P2 / H7  | $\forall K : \chi_{\text{BAO}}^2(\Omega_m=1, K) \geq 1415.82$                                             |        | <code>P2.EdS.chi2_slab</code>       | yes                  |
| P2 / H8  | preregistered slabs excluded at $\Delta\chi^2 > 25$                                                       |        | 19 slab files                       | <b>failed</b> (8/19) |
| P2 / A2  | $\forall \Omega_m \in [0, 0.255] \cup [0.35, 1], \forall K : \chi_{\text{BAO}}^2 > 35.2897$               |        | 29 slab theorems                    | yes (post hoc)       |
| P3 / H9  | $\chi_{\text{tot}}^2(0.29743, h_*, 0.02218) \leq 10.2897$ ,<br>$h_* \approx 0.685901$ (one-sided)         |        | <code>P3.Point.chi2tot_point</code> | yes                  |
| P3 / H10 | $\forall \Omega_m \in [0, 1], \forall \omega_b, \forall h \geq 0.73 : \chi_{\text{tot}}^2 > 10.2897 + 25$ |        | <code>P3.H0_ge_73_excluded</code>   | yes                  |
| P3 / H11 | the same for $h \leq 0.64$                                                                                |        | –                                   | not attempted        |

Table 1: Results (from `results/certified_numerics/P2_likelihood/result.json` and `P3_h0/result.json`).

### 3 Method

**Distances.** As in [5]:  $1/E$  is decreasing, so Riemann sums on a grid bound  $\chi$ ; an untrusted program supplies integer bounds of  $1/E$  at the grid nodes, which the kernel re-checks exactly (4661 nodes per parameter value, 46 601 for the fit point).

**Likelihood.**  $\chi^2$  is a quadratic form in the predictions. Writing  $K = K_0 + t$  for an untrusted centre  $K_0$ ,  $\chi^2 = c' - tB' + t^2A$  with  $c', B', A$  polynomials in the base quantities; a rational interval evaluator, proved sound once, bounds them over a box. Then  $A > 0$  gives  $\chi^2 \geq c' - B'^2/(4A)$  for *every* real  $K$ , which profiles over  $hr_d$  exactly. Centring (preregistration amendment A1, made before any number of this part was computed) keeps  $c'$  and  $B'$  small; without it, cancellation between two terms of size  $\sim 10^5$  would, by an a priori estimate, cost  $\sim 10^2$  and make the bounds vacuous.

**Parameter slabs.**  $\chi$ ,  $1/E$  and  $(z\chi^2/E)^{1/3}$  decrease in  $\Omega_m$ , so tables at the two edges of a slab bound every prediction inside it.

**The  $H_0$  statement.** On  $\{h : \omega_\nu \leq (1 - 2\alpha)\Omega_m h^2\}$ , which contains every box below at  $h \geq 0.73$ ,  $hr_d$  increases with  $h$  (imported theorem), and  $r_d$  decreases with  $\omega_{cb}$  and  $\omega_b$ . Hence over a box  $\Omega_m \in [\Omega_-, \Omega_+]$ ,  $\omega_b \in [\omega_-, \omega_+]$  every admissible  $K$  satisfies  $K \leq K_0 := c/(100 \cdot 0.73 r_d(\omega_{cb}(0.73, \Omega_+), \omega_+))$ . Centring at  $K_0$ , if  $B' \geq 0$  on the box then  $\chi^2(K) \geq c'$  for all  $K \leq K_0$ . The value of  $r_d$  at the box corner is enclosed through  $\exp(c_0 - \alpha \log \omega_{cb} - \beta \log \omega_b)$ , with log bounded by Mathlib’s series for  $\log(1 - u)$  after scaling by  $2^k$ , log 2 by Mathlib’s  $10^{-10}$  bracket, and exp by Mathlib’s Taylor

bound. At  $(\omega_{cb}, \omega_b) = (0.159228, 0.02218)$  the certified bracket is  $[143.10583303, 143.10583307]$  (relative width  $2.3 \times 10^{-10}$ ) and contains the Python value  $143.1058330475$ , within  $2.5 \times 10^{-12}$  of its midpoint. Outside  $\omega_b \in \mu \pm 7\sigma$  (including  $\omega_b \leq 0$ , where  $r_d$  is not physical) the prior term alone exceeds 49, using a kernel-checked proof that the BAO  $\chi^2$  is non-negative (block positive semidefiniteness of  $P$ ). Outside  $\Omega_m \in [0.255, 0.35]$  the all- $K$  slab theorems of (i) apply. The final theorem is assembled by nested case splits that replay the bisection of the  $(\Omega_m, \omega_b)$  band into 89 boxes.

## 4 Results

**(i) Matter density.** The interval  $[10.2525, 10.2897]$  for  $\chi^2$  at the fit point contains the value 10.27106 from the author’s own Rust BAO code (rusty-SUNDIALS, an unmerged branch; not independent of this work’s data handling) and a 40-digit value computed in an independent verification run. Einstein–de Sitter is excluded at every  $hr_d$ . The preregistered slab layout excluded only 8 of 19 slabs, so the preregistered prediction ( $[0, 0.24] \cup [0.36, 1]$  excluded) was not met: nine of the eleven failures are slack of the box bound in wide slabs, and the slabs  $[0.25, 0.26]$  and  $[0.34, 0.35]$  reach into the allowed region and cannot be excluded at all. A refinement by bisection, written after an untrusted preview and disclosed as amendment A2 (as the preregistration required for any change of layout), gives 29 slab theorems with bounds  $\geq 36.02$  covering  $[0, 0.255] \cup [0.35, 1]$ ; floating  $\Delta\chi^2 = 25$  edges are 0.2577 and 0.3443, so the certified bands are conservative. Controls: a copy of the data with one value shifted by  $10\sigma$  moves the certified interval to  $[122.69, 122.76]$ ; a slab containing the fit point is not excluded; false bounds fail to compile; planted `sorry` and axioms are flagged (the producer’s audit covered 59 theorems; the verification run re-audited a subset).

**(ii) Hubble constant.**  $\chi_{\text{tot}}^2 \leq 10.2897$  at  $(\Omega_m, h, \omega_b) = (0.29743, h_\star, 0.02218)$ ,  $h_\star = 1371802127/20000000000$ ; only this upper end was certified, which is all the exclusion needs (H9 met in that one-sided form). For all  $h \geq 0.73$ ,  $\chi_{\text{tot}}^2 > 10.2897 + 25$ ; the weakest element of the proof is the  $\Omega_m$  slab  $[0.2525, 0.255]$  at 36.02, the weakest  $H_0$  box clears at 36.82. With one parameter of interest and the others profiled,  $\Delta\chi^2 = 25$  corresponds to  $5\sigma$  (about  $4.3\sigma$  if read jointly in  $(\Omega_m, h, \omega_b)$ ). An independent verification run puts the floating  $\Delta\chi^2 = 25$  crossing at  $H_0 \approx 71.6$  and  $\Delta\chi^2 \approx 54.6$  at  $H_0 = 73$ ;  $H_0 = 73$  is a reference value, not the edge of the excluded region. That run also rebuilt all 173 modules of the final theorem’s import closure from source and found that the proof breaks when the threshold is raised to 65 (a false statement:  $\chi_{\text{tot}}^2 = 64.91$  at  $h = 0.73$ ), or when one box’s bound is raised or its  $K_0$  lowered beyond what its certificate supports (tightness checks; such a raised bound may itself be true). Planner-level checks (Python mirror, not Lean): for  $h \geq 0.686$ , and for  $h \geq 0.73$  with the BBN mean moved to 0.0282, the planner certified none of its 291 boxes; every failure was the vertex condition  $B' \geq 0$ , at boxes whose  $\chi^2$  lower bound was far above threshold, so these runs show that the method does not certify there, not that it detects a low  $\chi^2$ ; independent numerics show both statements are false. The lower side ( $h \leq 0.64$ ) was preregistered as secondary and not attempted.

## 5 Limitations and scope

Radiation is neglected in  $E(z)$  (it changes  $E(2.33)$  by about  $5 \times 10^{-4}$ ), the massive neutrino is treated as matter at all  $z$ , while the Aubourg fitting formula was calibrated on Boltzmann-code results that include radiation (with  $N_{\text{eff}} = 3.046$ ; DESI uses 3.044); the sound horizon is a fitting formula;  $\Omega_m$  is restricted to  $[0, 1]$ ; the likelihoods are Gaussian and the data are taken as published;

$\Delta\chi^2$  thresholds are conventions. The certified regions are conservative and depend on the box layout. Nothing here bears on the origin of the tension.

## Reproducibility and acknowledgements

Lean: `formal_cert/BA0Cert/` (Likelihood.lean, P2/, H0/, P3/); emitters, planners, evaluation and the client used to query the Rust code: `scripts/certified_numerics/`; preregistrations, amendments, compile logs, axiom audits, verification notes and results: `results/certified_numerics/P2_likelihood/` and `P3_h0/`. All are in the accompanying Zenodo bundle (with a manifest of sha256 sums) and in the AutoevolveAI repository, branch `worktree-openai-math-discovery`. Toolchain: Lean 4.34.1, Mathlib d13f23b7 [6, 7]. Numbers in this paper are generated from the result files by `papers/certified_desi_h0/gen_numbers.py`. Preregistration, Lean and Python code, numerics, the independent verification runs (separate instances of the same AI model, without access to the producer’s build outputs) and drafting were performed by the AI model Claude (Anthropic) under the author’s high-level direction; no human has independently re-derived the results; the author is responsible for the content.

## References

- [1] DESI Collaboration, DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints, arXiv:2503.14738 (2025).
- [2] N. Schöneberg, The 2024 BBN baryon abundance update, arXiv:2401.15054 (2024); A.-K. Burns, T. M. P. Tait and M. Valli, PRyMordial, Eur. Phys. J. C 84, 86 (2024).
- [3] É. Aubourg et al., Cosmological implications of baryon acoustic oscillation measurements, Phys. Rev. D 92, 123516 (2015), arXiv:1411.1074, eq. 16.
- [4] A. G. Riess et al., A comprehensive measurement of the local value of the Hubble constant, Astrophys. J. Lett. 934, L7 (2022), arXiv:2112.04510.
- [5] X. Callens, Kernel-checked enclosures of DESI DR2 BAO distances in Lean 4, Zenodo, doi:10.5281/zenodo.23247730 (2026).
- [6] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, CADE-28 (2021).
- [7] The mathlib Community, The Lean mathematical library, CPP 2020.